

Jour 2 : Protocole ARP et Gestion de Table (Article 1 - fin)

- **Pratique** : Décoder les entêtes Ethernet. Gestion des adresses MAC

Nous utilisons ce code qui va permettre de parser el contenu de la trame Ethernet et en utilisant la structure fourni dans le document sur [⊕ Let's code a TCP/IP stack, 1: Ethernet & ARP](#)
On peut obtenir un programme qui sépare les différents champs de la trame ethernet.

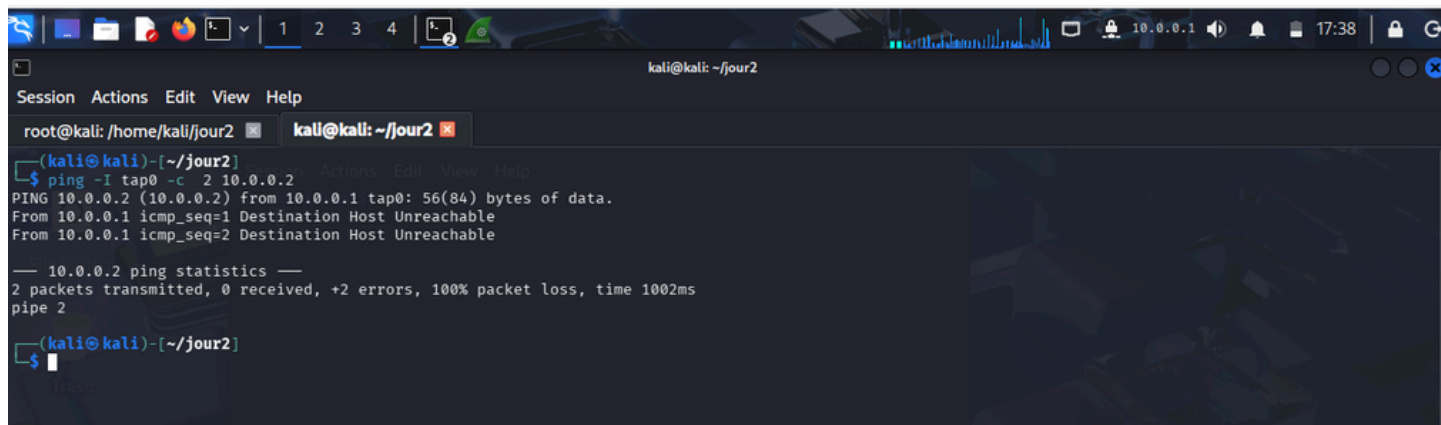
```
while (1) {
    int nread = read(tun_fd, buf, sizeof(buf));
    if (nread < 0) {
        perror("read");
        break;
    }
    if (nread < 14) continue; // Trop petit pour un header Ethernet

    struct eth_hdr *eth = (struct eth_hdr *)buf;

    // On convertit ethertype en little-endian pour l'afficher
    uint16_t type = ntohs(eth->ethertype);

    printf("\n=== Trame reçue (%d octets) ===\n", nread);
    printf("Dst MAC : %02x:%02x:%02x:%02x:%02x:%02x\n",
        eth->dmac[0], eth->dmac[1], eth->dmac[2],
        eth->dmac[3], eth->dmac[4], eth->dmac[5]);
    printf("Src MAC : %02x:%02x:%02x:%02x:%02x:%02x\n",
        eth->smac[0], eth->smac[1], eth->smac[2],
        eth->smac[3], eth->smac[4], eth->smac[5]);
    printf("EtherType : 0x%04x → ", type);

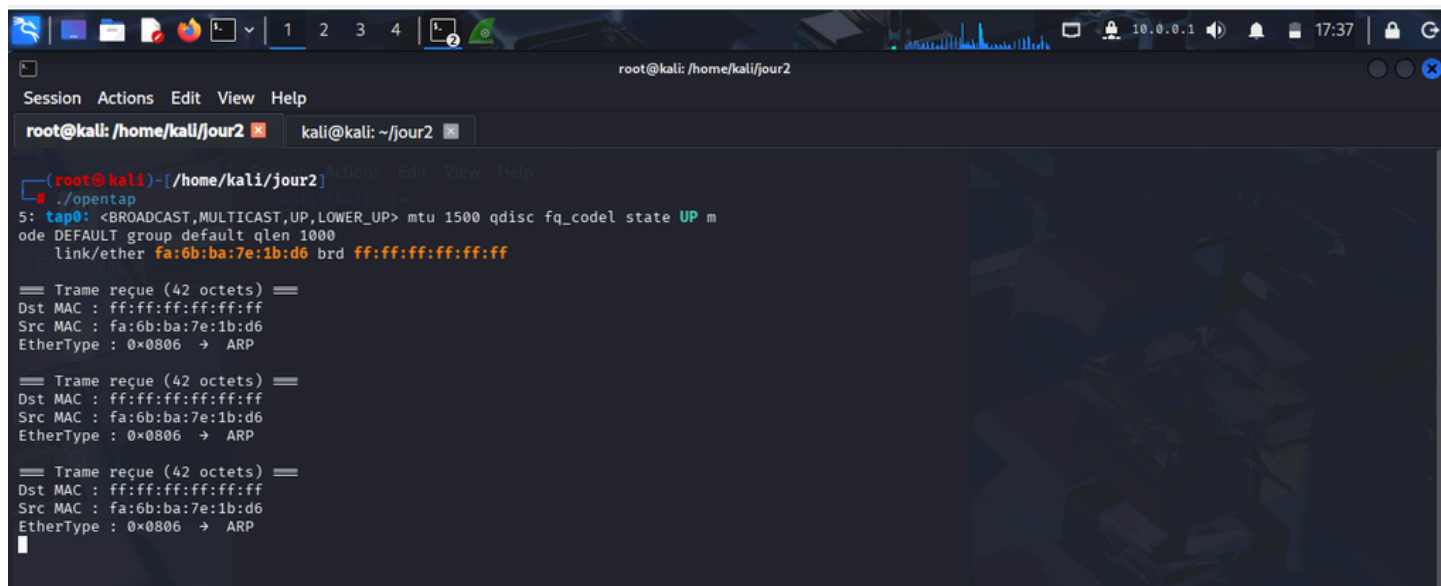
    switch (type) {
        case ETH_P_IP:    printf("IPv4\n");    break;
        case ETH_P_ARP:   printf("ARP\n");     break;
        case ETH_P_IPV6:  printf("IPv6\n");    break;
        default:          printf("Inconnu\n");  break;
    }
}
```



```
kali@kali: ~/jour2
Session Actions Edit View Help
root@kali: /home/kali/jour2 kali@kali: ~/jour2
(kali@kali)~[/jour2]
$ ping -I tap0 -c 2 10.0.0.2
PING 10.0.0.2 (10.0.0.2) from 10.0.0.1 tap0: 56(84) bytes of data.
From 10.0.0.1 icmp_seq=1 Destination Host Unreachable
From 10.0.0.1 icmp_seq=2 Destination Host Unreachable

--- 10.0.0.2 ping statistics ---
2 packets transmitted, 0 received, +2 errors, 100% packet loss, time 1002ms
pipe 2

(kali@kali)~[/jour2]
$
```



```
root@kali: /home/kali/jour2
Session Actions Edit View Help
root@kali: /home/kali/jour2 kali@kali: ~/jour2
(kali@kali)~[/home/kali/jour2]
$ ./opentap
5: tap0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP m
ode DEFAULT group default qlen 1000
    link/ether fa:6b:ba:7e:1b:d6 brd ff:ff:ff:ff:ff:ff

== Trame reçue (42 octets) ==
Dst MAC : ff:ff:ff:ff:ff:ff
Src MAC : fa:6b:ba:7e:1b:d6
EtherType : 0x0806 -> ARP

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C'est mieux maintenant , on voit la trame de manière lisible et compréhensible au lieu d'une suite de bit sur wireshark.

- **Focus Sécurité** : MAC Spoofing. Si je change un octet dans cet entête, je change mon identité physique sur le réseau.